

Problems (Advanced Control)

[Lyapunov Functions and Stability]

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1. For what values of ω is the discrete-time system

$$x(k+1) = \begin{pmatrix} \frac{1}{2} & \omega \\ -\omega & \frac{1}{2} \end{pmatrix} x(k)$$

globally asymptotically stable? When $\omega = \frac{1}{2}$, let P be the solution to the equation

$$A^T P A - P = -I.$$

Is it the case that $P \succ 0$? Did you need to compute P ?

$$A = \begin{bmatrix} \frac{1}{2} & \omega \\ -\omega & \frac{1}{2} \end{bmatrix}$$

$$\therefore \text{for } \lambda, \det \begin{bmatrix} \frac{1}{2} - \lambda & \omega \\ -\omega & \frac{1}{2} - \lambda \end{bmatrix} = 0$$

$$\therefore \left(\frac{1}{2} - \lambda\right)^2 + \omega^2 = \lambda^2 - \lambda + \left(\frac{1}{4} + \omega^2\right) = 0$$

$$\therefore \lambda = \frac{1 \pm \sqrt{1 - (1 + 4\omega^2)}}{2} = \frac{1}{2} \pm \omega i$$

$$\therefore |\lambda| = \sqrt{\frac{1}{4} + \omega^2} < 1 \text{ for DT globally stable}$$

$$\therefore \omega^2 < \frac{3}{4}$$

$$\therefore -\frac{\sqrt{3}}{2} < \omega < \frac{\sqrt{3}}{2}$$

$$\text{when } \omega = \frac{1}{2},$$

$$A = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

since DT system is globally asymptotically stable, and

$$A^T P A - P = -I$$

$$\therefore P > 0$$

This can be prove by let $P = Q + A^T Q A + (A^T)^2 Q A^2 + \dots = \sum_{i=0}^{\infty} (A^T)^i Q A^i$

$$\therefore A^T P A = \sum_{i=1}^{\infty} (A^T)^i Q A^i = -Q + \sum_{i=0}^{\infty} (A^T)^i Q A^i = -Q + P$$

2. Consider a nonlinear discrete time system

$$x(k+1) = f(x(k))$$

with equilibrium point $x_e \neq 0$. Let $g(x) = f(x + x_e) - x_e$ and show that:

(i) The system can be written in the equivalent form

$$\tilde{x}(k+1) = g(\tilde{x}(k)) \quad (*)$$

where $\tilde{x}(k) = x(k) - x_e$.

(ii) $x = 0$ is an equilibrium point of the above system.

This implies that stability of the original system about x_e can be tested by finding a Lyapunov function for the system (*).

$$(i) \quad \tilde{x}(k+1) = x(k+1) - x_e$$

$$= f(x(k)) - x_e$$

$$= f(x(k) - x_e + x_e) - x_e$$

$$= f(\tilde{x}(k) + x_e) - x_e$$

$$= g(\tilde{x}(k))$$

(ii) At equilibrium point,

$$\tilde{x}_e = x_e - x_e = 0$$

$$g(\tilde{0}) = f(-x_e + x_e) - x_e = -x_e = \tilde{x}(0)$$

$\therefore$ At equilibrium point, $\dot{x}_e = f(x_e) = x_e$

$\therefore \tilde{x}_e$ is at equilibrium point

3. Use the Lyapunov function $V(x) = \|x\|^2$ to show that the following systems are asymptotically stable

(i) The one-dimensional system $\dot{x} = -x^3$.

(ii) The 2-dimensional system

$$\dot{x}_1 = -x_2 - x_1(1 - x_1^2 - x_2^2)$$

$$\dot{x}_2 = x_1 - x_2(1 - x_1^2 - x_2^2).$$

Which of these systems (if any) is globally asymptotically stable?

$$\phi \quad V(x) = \|x\|^2 = x^2$$

$$\therefore V(0) = 0 \text{ and } V(x) > 0 \text{ for any } x \neq 0 \quad (L1)$$

$$(\nabla V \cdot f)(x) = \frac{\partial V}{\partial x} \cdot \frac{\partial x}{\partial t} = 2x \cdot (-x^3) = -2x^4$$

$$\therefore (\nabla V \cdot f)(x) < 0 \text{ for } x \neq 0 \quad (L2)$$

In addition, $V(x) \rightarrow \infty$ as $\|x\| \rightarrow \infty$

$\therefore$ the system is globally asymptotically stable

$$(ii) \quad V(x) = \|x\|^2 = x_1^2 + x_2^2$$

$$\therefore V(0) = 0 \text{ and } V(x) > 0 \text{ for } x \neq 0 \quad (L1)$$

$$\nabla V(x) = \begin{bmatrix} 2x_1 \\ 2x_2 \end{bmatrix}$$

$$f(x) = \begin{bmatrix} -x_2 - x_1(1 - x_1^2 - x_2^2) \\ x_1 - x_2(1 - x_1^2 - x_2^2) \end{bmatrix}$$

$$\begin{aligned} \therefore (\nabla V \cdot f)(x) &= -\frac{2}{\lambda}(x_1^2 + x_2^2)[1 - (x_1^2 + x_2^2)] \\ &= -\|x\|^{\frac{2}{\lambda}}(1 - \|x\|^{\frac{2}{\lambda}}) \end{aligned}$$

$$\therefore (\nabla V \cdot f)(x) < 0 \text{ for any } \|x\| < 1 \quad (L2)$$

$\therefore$ the system is locally asymptotically stable

4. Show that the nonlinear system

$$\dot{x}_1 = -x_1 + x_1x_2$$

$$\dot{x}_2 = -x_2$$

is globally asymptotically stable by analyzing the function $V(x) = \ln(1 + x_1^2) + x_2^2$.

$$V(0) = \ln(1) = 0, \quad V(x) \rightarrow \infty \quad \text{as} \quad \|x\| \rightarrow \infty \quad (L1)$$

$$\nabla V(x) = \begin{bmatrix} \frac{2x_1}{1+x_1^2} \\ 2x_2 \end{bmatrix}$$

$$\begin{aligned} \therefore (\nabla V \cdot f)(x) &= \frac{2x_1^2(-1+x_2)}{1+x_1^2} - 2x_2^2 \\ &= \frac{-2x_1^2 + 2x_1^2x_2 - 2x_2^2 - 2x_1^2x_2^2}{1+x_1^2} \\ &= \frac{-2x_1^2 - 2x_2^2 - 2x_1^2(x_2^2 - x_2)}{1+x_1^2} \\ &= -2 \frac{x_1^2(x_2^2 - x_2 + 1) + x_2^2}{1+x_1^2} \\ &= -2 \frac{x_1^2(x_2^2 - x_2 + \frac{1}{4} + \frac{3}{4}) + x_2^2}{1+x_1^2} \\ &= -2 \frac{x_1^2[(x_2 - \frac{1}{2})^2 + \frac{3}{4}] + x_2^2}{1+x_1^2} \\ &< 0 \quad \text{for } x \neq 0 \quad (L2) \end{aligned}$$

5. The Euler angles $\omega = (\omega_1 \ \omega_2 \ \omega_3)^\top$ for the rotation of a satellite satisfy the differential equation

$$J_1 \dot{\omega}_1 = (J_2 - J_3) \omega_2 \omega_3 + u_1$$

$$J_2 \dot{\omega}_2 = (J_3 - J_1) \omega_3 \omega_1 + u_2$$

$$J_3 \dot{\omega}_3 = (J_1 - J_2) \omega_1 \omega_2 + u_3$$

where J_i are constants corresponding to the moments of inertia, and u_i are control inputs. Show that the feedback law $u_i = -k_i \omega_i$ makes the system globally asymptotically stable if $k_1, k_2, k_3 > 0$.

$$\text{Let } V(\omega) = \frac{1}{2} (J_1 \omega_1^2 + J_2 \omega_2^2 + J_3 \omega_3^2)$$

$$\therefore V(0) = 0 \text{ and } V(\omega) \rightarrow \infty \text{ as } \|\omega\| \rightarrow \infty \quad (1)$$

$$\begin{aligned} \therefore (\nabla V \cdot f)(\omega) &= (J_2 - J_3) \omega_1 \omega_2 \omega_3 - k_1 \omega_1^2 \\ &\quad + (J_3 - J_1) \omega_1 \omega_2 \omega_3 - k_2 \omega_2^2 \\ &\quad + (J_1 - J_2) \omega_1 \omega_2 \omega_3 - k_3 \omega_3^2 \\ &= -k_1 \omega_1^2 - k_2 \omega_2^2 - k_3 \omega_3^2 \end{aligned}$$

$$\therefore (\nabla V \cdot f)(\omega) < 0 \text{ for any } \omega \neq 0 \quad (2)$$

6. Consider the nonlinear system

$$\begin{aligned}\dot{x}_1 &= -x_2 - \frac{3}{2}x_1^2 - \frac{1}{2}x_1^3 \\ \dot{x}_2 &= 3x_1 - x_2\end{aligned}$$

from Example 4.15 in lectures.

(i) Show that the system is (locally) asymptotically stable about the equilibrium point $x = 0$.

(ii) Find the matrix

$$A = \frac{\partial f}{\partial x}(0)$$

which describes the linearized dynamics of the system.

(iii) Find a matrix P such that $PA + A^T P = -I$. Why must P be strictly positive definite?

(iv) For what values of $r > 0$ is $\Omega_r = \{x : x^T P x \leq r\}$ a region of attraction? You may need to explore this sub-question using computational tools.

(iii) Let $V(t) = \frac{1}{2}(x_1^2(t) + x_2^2(t))$. Show that there exists $c > 0$ such that

$$\frac{dV}{dt} \leq -cV(t) \rightarrow \text{system bounded}$$

whenever $\|x(t)\|^2 \geq 10$. What does this mean for the trajectories of the system if the initial condition $x(0)$ is large? *system will enter Ω_r some t for any x .*

(i) At $x = 0$,

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

(ii)

$$A = \frac{\partial f}{\partial x}(0) = \begin{bmatrix} -3x_1 - \frac{3}{2}x_1^2 & -1 \\ 3 & -1 \end{bmatrix}_{x_1, x_2 = 0} = \begin{bmatrix} 0 & -1 \\ 3 & -1 \end{bmatrix}$$

iii

$$\text{Let } P = \begin{bmatrix} p & q \\ q & r \end{bmatrix}$$

$$\begin{aligned} \therefore \begin{bmatrix} p & q \\ q & r \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 3 & -1 \end{bmatrix} + \begin{bmatrix} 0 & -1 \\ 3 & -1 \end{bmatrix}^T \begin{bmatrix} p & q \\ q & r \end{bmatrix} \\ = \begin{bmatrix} p & q \\ q & r \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 3 & -1 \end{bmatrix} + \begin{bmatrix} 0 & 3 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} p & q \\ q & r \end{bmatrix} \\ = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \end{aligned}$$

$$\therefore \begin{bmatrix} 6q & 3r - p - q \\ 3r - p - q & -2q - 2r \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\therefore q = -\frac{1}{6}, r = \frac{2}{3}, p = \frac{13}{6}$$

$$\therefore P = \begin{bmatrix} \frac{13}{6} & -\frac{1}{6} \\ -\frac{1}{6} & \frac{2}{3} \end{bmatrix}$$

$$\therefore p, r > 0 \text{ and } \det(P) = \frac{17}{12} > 0$$

$$\therefore P > 0$$

P must be positive definite so system is locally asymptotically stable

iv) see MATLAB

$$r = 5.5101 \times 10^{-7}$$

$$(v) \frac{dV}{dt} = (\nabla V \cdot f)(x)$$

$$= -x_1 x_2 - \frac{3}{2} x_1^3 - \frac{1}{2} x_1^4 + 3x_1 x_2 - x_2^4$$

$$= -\frac{1}{2} x_1^4 - \frac{3}{2} x_1^3 + \underbrace{2x_1 x_2}_{\text{dominates near 0}} - x_2^4 \leq -\frac{1}{2} x_1^4 - \frac{3}{2} x_1^3 + (x_1^2 + x_2^2) - x_2^4$$

dominates near 0

$$= -\frac{1}{2} x_1^4 (x_1^2 + 3x_1 - 2)$$

$$< 0 \text{ when } x_1 > -\frac{3-\sqrt{17}}{2} \text{ or } x_1 < -\frac{3+\sqrt{17}}{2}$$

$$(iv) \text{ let } V(x) = x^T P x,$$

$$(\nabla V \cdot f)(x) = 2x^T P \cdot [Ax + g(x)]$$

$$= 2x^T P A x + 2x^T P g(x)$$

$$= x^T (A^T P + P A) x + 2x^T P g(x)$$

$$= x^T (-I) x + 2x^T P g(x)$$

$$= -\|x\|^2 + 2x^T P g(x)$$

$$= \underbrace{-x_1^2 - x_2^2}_{\text{dominates near 0}} - \frac{13}{6} x_1^3 + \frac{1}{6} x_1^3 x_2 - \frac{13}{2} x_1^3 + \frac{1}{2} x_1^3 x_2$$

dominates near 0

$$\text{if } (\nabla V \cdot f)(x) < 0 \text{ for all } x \in \Omega_r = \{x : V(x) \leq r\}$$

the level set Ω_r is a region of attraction $\rightarrow$ see solution

7. Let $V : \mathbb{R}^n \rightarrow \mathbb{R}$ satisfy $V(x) > 0$ and $V(x) \rightarrow \infty$ as $\|x\| \rightarrow \infty$. Show that any level set $\Omega_c = \{x : V(x) \leq c\}$ satisfies

$$B_{R_1} \subset \Omega_c \subset B_{R_2}$$

for some $0 < R_1 < R_2$, where $B_R = \{x : \|x\| \leq R\}$ is the ball of radius R . This shows that each level set of V is a bounded subset of $\mathbb{R}^n$.

- if $c > 0$, it follows fact that $V(0) = 0$ and that V is continuous at 0, there is a ball B_{R_1} contained in Ω_c .
- let $z \in \Omega_c$, then $V(z) \in C$, since $V(x) \rightarrow \infty$ when $x \rightarrow \infty$, there exists $R_2 > 0$ such that $V(x) > c$ whenever $\|x\| > R_2$, therefore $\|z\| \leq R_2 < \|x\|$, so $z \in B_{R_2}$, $\Omega_c \subset B_{R_2}$.

8. This question fills in the details of why the conditions for global stability of nonlinear system $x(k+1) = f(x(k))$ work.

Suppose that

$$\alpha_1 \|x\|^2 \leq V(x) \leq \alpha_2 \|x\|^2$$

and that the strict decay condition $V(f(x)) < V(x)$ holds for any $x \in \mathbb{R}^n$.

- (i) Show that the sequence $a_n = V(x_n)$ is decreasing that that $a_n \geq 0$ for all n .
- (ii) Conclude that the series converges to a limit $a_n \rightarrow a$ as $n \rightarrow \infty$ and that $a \geq 0$.
- (iii) Now let R_1, R_2 be such that $B_{R_1} \subset \Omega_a \subset B_{R_2}$ (see Problem 7). Define

$$\gamma = \min_{R_1 \leq \|x\| \leq R_2} \{V(f(x)) - V(x)\} \neq 0.$$

Think about what happens if $V(x_n) \rightarrow a$ and $a > 0$. This could be helped by drawing a sketch containing the origin, the balls B_{R_1}, B_{R_2} , a level set Ω_a for $a > 0$, and a trajectory converging to the boundary of the level set when $V(x) = a$. Explain why it must actually be the case that $a = 0$.

- (iv) Show the $x(k+1) = f(x(k))$ is globally asymptotically stable.

$$(i) \quad a_{n+1} = V(x_{n+1}) = V(f(x_n)) < V(x_n) = a_n$$

$$\therefore a_{n+1} < a_n \text{ for all } n$$

$$\therefore V(x) \geq 0$$

$$\therefore 0 \leq a_{n+1} < a_n \text{ for all } n$$

(ii) the sequence $(a_n)_{n=0}^{\infty}$ is decreasing and bounded by a ,

where $a \geq 0$ since $0 \leq a_{n+1} < a_n$ for all n

* if $a = 0$, $V(x_n) = a_n \rightarrow 0$ for $n \rightarrow \infty$, therefore

$$\alpha_1 \|x_n\|^2 \leq V(x_n) \rightarrow 0, \text{ so } x_n \rightarrow 0 \text{ for } n \rightarrow \infty,$$

so $x_n = f(x_n) = 0$ for $n \rightarrow \infty$, globally stable

(iii) see solution

(iv) see solution

9. Let $P \succ 0$ be a symmetric and strictly positive definite matrix. Show that all the diagonal elements of P must satisfy $p_{ii} > 0$. Explain why $\det P > 0$.

• let $e_i = [0 \ 0 \ \dots \ 0 \ 1 \ 0 \ \dots \ 0 \ 0]^T$ where 1 at $(i, 1)$,

then $e_i^T P e_i = p_{ii}$

so if $P \succ 0$, $p_{ii} > 0$ for $i = 1, \dots, n$

• all $\lambda(P) > 0$ for $P \succ 0$

$\therefore \det(P) = \text{product of all } \lambda(P) > 0$

10. Which of the following matrices are strictly positive definite?

$$P_1 = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \quad P_2 = \begin{pmatrix} 100 & 1 & 5 & -7 \\ 1 & 20 & 0 & -3 \\ 5 & 0 & -1 & 4 \\ -7 & -3 & 4 & 100 \end{pmatrix}, \quad P_3 = \begin{pmatrix} 1 & 0.1 & 0 \\ 0.1 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

You should not need to compute the eigenvalues!

for P_1 , $p_{ii} > 0$, $\det(P_1) = 1 \cdot (0 - 1) - 0 + 1 \cdot (1 - 4) = -4 < 0$

$\therefore P_1$ is ^{not} strictly positive definite

for P_2 , $p_{33} < 0$

$\therefore P_2$ is not positive definite

for P_3 , $p_{ii} > 0$, $\det(P_3) = 0 - 0 + 1 \cdot (2 - 0.01) = 1.99$

$\therefore P_3$ is ^{strictly} not positive definite

11. Suppose that $P \succ 0$ is symmetric and positive definite. Let $V(x) = x^\top Px$. Show that

$$(\nabla V)(x) = \begin{pmatrix} \frac{\partial V}{\partial x_1} \\ \vdots \\ \frac{\partial V}{\partial x_n} \end{pmatrix} = 2Px$$

Also show that

$$\nabla V(x) \cdot Ax = 2x^\top PAx = x^\top (PA + A^\top P)x$$

for any vector x and matrix $A \in \mathbb{R}^{n \times n}$.

Assume $(\nabla V)(x) = 2Px$.

$$\nabla V(x) \cdot Ax = 2Px \cdot Ax = 2x^T P^T Ax$$

$$\therefore P = P^T$$

$$(Px) \cdot (Ax) = (Ax) \cdot (Px) = x^T P Ax + x^T A^T P x$$

$$\begin{aligned}\therefore \nabla V(x) \cdot Ax &= 2x^T P Ax = x^T P Ax + x^T A P x \\ &= x^T (PA + AP) x\end{aligned}$$

for $(\nabla V)(x)$.

$$V(x) = x^T P x = \sum_{i,j=1}^n x_i P_{ij} x_j$$

$$\therefore \frac{\partial V}{\partial x_k} = \sum_{i=1}^n x_i P_{ik} + \sum_{j=1}^n P_{kj} x_j$$

$$= 2(Px)_k \text{ where each } k = 1, \dots, n$$

$$\text{Therefore, } (\nabla V)(x) = \begin{bmatrix} \frac{\partial V}{\partial x_1} \\ \frac{\partial V}{\partial x_2} \\ \dots \\ \frac{\partial V}{\partial x_n} \end{bmatrix} = \sum_{k=1}^n 2(Px)_k = 2Px$$